








Advanced Time Series Analysis and Forecasting

Chapter 3: Structural VAR and local projections



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Acronyms used in this material (1/2)

- **AB**: the AB model of a structural VAR (A times residuals = B times shocks)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **ARDL**: AutoRegressive Distributed Lag (model)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BQ**: Blanchard–Quah (long-run identification)
- **CEE**: Christiano–Eichenbaum–Evans (recursive monetary VAR, 1999)
- **CPI**: Consumer Price Index
- **DGP**: Data-Generating Process
- **EA**: Euro Area
- **ECB**: European Central Bank
- **EIA**: US Energy Information Administration
- **FEVD**: Forecast Error Variance Decomposition
- **FF4**: surprise in the federal funds futures contract three months ahead
- **FOMC**: Federal Open Market Committee
- **FRA**: Forward Rate Agreement
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product

Acronyms used in this material (2/2)

- **GK**: Gertler–Karadi (proxy SVAR, 2015)
- **GNP**: Gross National Product
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HICP**: Harmonised Index of Consumer Prices
- **HQ**: Hannan–Quinn (information criterion)
- **IP**: Industrial Production
- **LLM**: Large Language Model
- **LP**: Local Projection
- **LP-IV**: Local Projection with Instrumental Variables
- **MA**: Moving Average
- **MSE**: Mean Squared Error
- **OLS**: Ordinary Least Squares
- **QR**: Quick Response (code)
- **RMSE**: Root Mean Squared Error
- **ROBOR**: Romanian Interbank Offer Rate
- **RZ**: Ramey–Zubairy (state-dependent multipliers, 2018)
- **SCA**: Seasonally and Calendar Adjusted
- **SE**: Standard Error
- **SVAR**: Structural Vector AutoRegression
- **TSA**: Time Series Analysis
- **US**: United States
- **VAR**: Vector AutoRegression
- **VECM**: Vector Error Correction Model

Today's question and route

- **Question**: what does a monetary policy shock, an oil supply shock or a fiscal shock *cause*, and how sure can we be?
  - a reduced-form VAR describes correlations; a causal answer needs an identifying assumption, stated and defended
- **Route** of the chapter
  - the identification problem; short-run (recursive and non-recursive) and long-run restrictions
  - sign, narrative and heteroskedasticity-based identification; set identification
  - external instruments and high-frequency surprises; inference for impulse responses
  - local projections: LP against VAR, LP-IV, state dependence; Romania and euro-area spillovers
- We build on TSA, Chapter 6 (reduced-form VAR, Granger causality, Cholesky responses, FEVD) and on Chapter 0 (HAC, bootstrap); Seminar 3 comes before this lecture

Learning outcomes

- State the identification problem of a structural VAR, count the restrictions it needs and explain observational equivalence through rotations
- Identify shocks by short-run, long-run, sign, narrative and heteroskedasticity restrictions, and say what each assumes about the economy
- Estimate a proxy SVAR with an external instrument, check its relevance and use inference that stays valid with weak instruments
- Build bootstrap bands for impulse responses (residual, wild, block; Kilian bias correction) and read them correctly
- Estimate local projections, LP-IV and state-dependent LP with HAC or lag-augmented inference, and choose between LP and VAR by the bias–variance trade-off

Reading, data and tools

- Backbone: Kilian and Lütkepohl (2017), Ch. 8–15 (identification, inference, LP); Hamilton (1994), Ch. 11; Lütkepohl (2005)
  - surveys: Ramey (2016) (shocks and their propagation), Stock and Watson (2018) (external instruments), Stock and Watson (2001)
- Python Quantlets of this chapter: Quantlets/Ch_03
  - VAR, Cholesky, long-run and sign identification, proxy SVAR, bootstraps, LP, LP-IV written out in `numpy`
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source	Use
US monthly macro data, FF4 surprises, Romer–Romer shocks	replication files of Ramey (2016) (public)	CEE, sign, proxy, LP
world crude oil production; refiner acquisition cost	EIA (bulk file, Monthly Energy Review)	Kilian
global real activity index; US CPI; real GNP; unemployment of men 20+	FRED (IGREA, CPIAUCSL, GNPC96, LNS14000025)	Kilian, BQ
US quarterly fiscal data 1889–2015	replication files of Ramey and Zubairy (2018)	state-dependent LP
IP, HICP, 3-month rates (RO, EA); EUR/RON	Eurostat; BNR reference rate	Romanian VAR
ECB monetary policy shocks; EA 1-year rate	Jarociński and Karadi (2020) (authors' update); ECB Data Portal	spillovers

- All sources are public and need no account or key; monthly data up to mid-2026, except the replication files
- IP: industrial production; HICP: harmonised index of consumer prices; FF4: surprise in the fourth federal funds futures contract

From correlations to shocks



Christopher Sims, Nobel lecture, 2011

Photo: Holger Motzkau (2011); CC BY-SA 3.0; Wikimedia Commons

- 1980: Sims (1980) proposes VARs instead of large models with "incredible" exclusion restrictions
  - the Cholesky ordering was his first identification scheme, and the first target of criticism
- 1989: long-run restrictions Blanchard and Quah (1989); 1999: the recursive monetary VAR Christiano, Eichenbaum and Evans (1999)
- 2005: sign restrictions Uhlig (2005) and local projections Jordà (2005); 2012–2018: external instruments Stock and Watson (2012), Mertens and Ravn (2013), Gertler and Karadi (2015)
- 2011: Sims and Sargent share the Nobel Prize "for their empirical research on cause and effect in the macroeconomy"
- The question has stayed the same; the identifying assumptions have become weaker and more transparent

Reduced form and structural form (1/2)

- Reduced form (TSA, Chapter 6): $y_t = \nu + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t$, $\mathbb{E} u_t u_t' = \Sigma_u$
  - y_t: the vector of n variables; ν: the intercepts; A_i: $n \times n$ coefficient matrices; Σ_u: the covariance matrix of u_t
  - u_t are one-step forecast errors: they mix all the economic shocks that hit in period t
- Structural form: $u_t = B_0 \varepsilon_t$, with structural shocks ε_t mutually uncorrelated, $\mathbb{E} \varepsilon_t \varepsilon_t' = I_n$
  - B_0: the $n \times n$ impact matrix, column j = the impact effect of shock j; I_n: the identity matrix (shocks of unit variance)
  - a **shock** is primitive, exogenous, uncorrelated with other shocks and economically meaningful Ramey (2016)

Reduced form and structural form (2/2)

- Equivalent structural equations: multiply by B_0^{-1}
  - $B_0^{-1} y_t = B_0^{-1}\nu + B_0^{-1}A_1 y_{t-1} + \cdots + \varepsilon_t$
  - the off-diagonal elements of B_0^{-1} are the contemporaneous relations among the variables
- Structural impulse responses: $\Theta_h = \Phi_h B_0$
  - Φ_h: the reduced-form MA coefficients, $y_t = \mu + \sum_h \Phi_h u_{t-h}$, $\Phi_0 = I_n$
  - element (i, j) of Θ_h: effect of a one-standard-deviation shock j on variable i after h periods

Counting restrictions: the order condition

- The data identify $\Sigma_u = B_0 B_0'$: $n(n+1)/2$ distinct moments; B_0 has n^2 unknowns
  - n: the number of variables; Σ_u is symmetric, so it has $n(n+1)/2$ distinct elements
  - order condition: at least $n(n-1)/2$ restrictions (3 for $n=3$, 28 for $n=8$)
  - the rank condition must also hold locally Rubio-Ramírez, Waggoner and Zha (2010): counting is necessary, not sufficient
- **Observational equivalence**: for any orthogonal Q ($QQ' = I$), $\tilde{B}_0 = B_0 Q$ gives $\tilde{B}_0 \tilde{B}_0' = \Sigma_u$
  - the two models fit the data equally well and imply different impulse responses
  - identification = a rule that selects one Q (point identification) or a set of Q (set identification)
- A useful parametrisation: $B_0 = PQ$ with P the Cholesky factor of Σ_u
  - P: the lower-triangular matrix with $PP' = \Sigma_u$; every identification scheme is a choice of Q

Invertibility and the limits of a VAR

- A VAR recovers ε_t from current and past y_t only if the shocks are **invertible** (fundamental)
  - news shocks and fiscal foresight break invertibility: agents react before the econometrician sees the shock
  - remedy: add forward-looking variables (asset prices, forecasts) or use the shock directly Ramey (2016)
- With an external instrument we need only *partial* invertibility of the shock of interest Stock and Watson (2018)
  - local projections with the shock as regressor do not need invertibility at all Plagborg-Møller and Wolf (2021)
- Practical rule: a small VAR omitting information the agents use (expectations, commodity prices) produces "puzzles"

Recap: The identification problem

- Σ_u fixes B_0 only up to an orthogonal rotation Q
- Point identification needs at least $n(n-1)/2$ restrictions and the rank condition
- Every scheme is an economic assumption: state it, defend it, test what can be tested
- Invertibility is a separate assumption; instruments and LP relax it

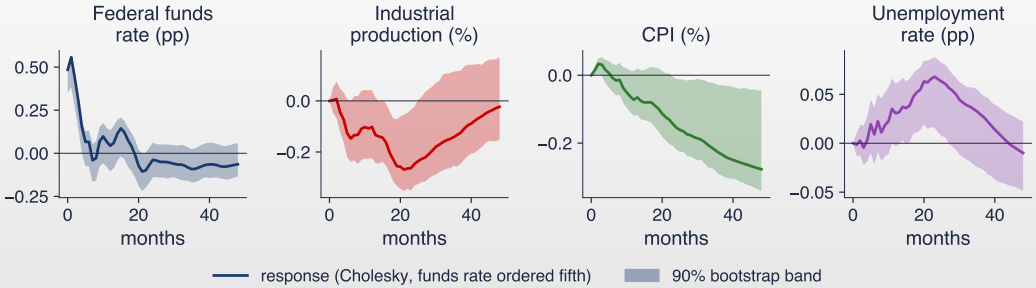
Recursive identification

- $B_0 = P$ lower triangular (zeros above the diagonal): variable i does not respond within the period to shocks ordered after it
  - exactly $n(n-1)/2$ zeros: just identified; the ordering is the assumption (TSA, Chapter 6 showed that it matters)
- Monetary block of Christiano, Eichenbaum and Evans (1999): slow variables (output, prices), the policy rate, fast variables (reserves, money)
  - assumption: within a month, output and prices do not react to policy, and policy reacts to them
  - only the position of the policy variable matters for its shock; the ordering inside the blocks is irrelevant
- Plausible at a monthly frequency with slow-moving real variables; implausible for asset prices or for quarterly data

Case study: Christiano, Eichenbaum and Evans (1999)

- The paper: the benchmark recursive monetary VAR; US monthly data, the federal funds rate shock ordered after output, prices and commodity prices Christiano, Eichenbaum and Evans (1999)
  - we use the specification of Ramey (2016), Figure 1: VAR(12) in log IP, unemployment, log CPI, log commodity prices, funds rate, log nonborrowed and total reserves, log M1
- Sample 1965:1–1995:6 ($T = 354$), constant, Cholesky with the funds rate fifth; 90% bands from 500 residual-bootstrap replications
  - the largest root of the companion matrix is 0.9999: a VAR in levels, consistent for responses at short and medium horizons Kilian and Lütkepohl (2017)
- Question: how do output and prices respond to an unexpected tightening?

A recursive monetary policy shock (CEE specification)



- Responses to a one-standard-deviation funds-rate shock (0.48 pp); 90% residual-bootstrap bands

ATS_ch3_recursive_var

Interpreting the CEE responses

- Industrial production falls slowly: trough -0.27% after 21 months (band $[-0.35, -0.05]$); unemployment rises by up to 0.07 pp
  - the hump shape and the delay are the classic "long and variable lags" of monetary policy
- Prices first rise slightly (0.033% after 2 months), then fall to -0.28% after four years
  - the initial rise is the **price puzzle**: the Fed tightens when it expects inflation; commodity prices in the VAR reduce but do not remove it
- Monetary shocks explain little of output: 6.6% of the 24-month forecast error variance of IP

Non-recursive short-run restrictions

- **AB model** Lütkepohl (2005), Kilian and Lütkepohl (2017): $Au_t = B\varepsilon_t$; zeros and known values anywhere in A and B
  - A: the contemporaneous relations among u_t; B: the impact of the shocks; then $\Sigma_u = A^{-1}BB'A^{-1\prime}$
  - estimated by maximum likelihood: maximise $-\frac{T}{2}\ln|\Sigma_u(A,B)| - \frac{T}{2}\mathrm{tr}[\Sigma_u(A,B)^{-1}\hat{\Sigma}_u]$
  - $|\cdot|$: the determinant; tr: the trace; $\hat{\Sigma}_u$: the residual covariance matrix of the reduced form
  - over-identifying restrictions can be tested by a likelihood-ratio test against the just-identified model
- **Blanchard–Perotti** Blanchard and Perotti (2002): the automatic response of taxes to output is an *external* elasticity (from tax codes)
  - with that elasticity fixed, the cyclically adjusted tax innovation is an instrument for the remaining coefficients
  - government spending is predetermined within the quarter (decision lags): a zero restriction with an institutional argument
- Seminar 3, A2: one known elasticity identifies a 2×2 system by instrumental variables

Limits of the recursive assumption

- Simultaneity: interest rates and asset prices react to each other within the day
  - no ordering of stock prices and the policy rate is credible at a monthly frequency
- Omitted information: the policy shock absorbs the Fed's forecasts
  - narrative shocks Romer and Romer (2004) purge the intended rate change of the Greenbook forecasts
- Sample dependence: the post-1983 responses are weaker and often puzzling Ramey (2016)
  - the remedies of the next sections: weaker restrictions (signs) or outside information (instruments)

Recap: Short-run restrictions

- Cholesky = timing assumptions; only the position of the shock variable matters for its responses
- CEE: output falls with a lag of about 21 months; the price puzzle remains
- Non-recursive schemes use institutional knowledge (tax elasticities, decision lags) and can be over-identified and tested
- Short-run zeros are least credible exactly where they matter most: fast markets

Not all oil price shocks are alike

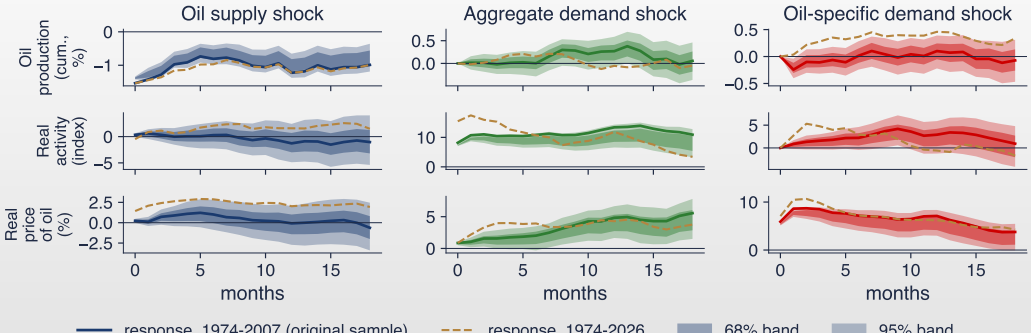


Burning Kuwaiti oil field, March 1991

Photo: JO1 Gawlowicz, US Navy (1991); public domain; Wikimedia Commons

- Kilian (2009): monthly VAR(24) in (i) the percent change of world crude oil production, (ii) global real activity, (iii) the log real price of oil
  - recursive: oil supply does not respond within the month to demand (adjustment costs); real activity does not respond within the month to oil-specific demand
- Three shocks: oil supply, aggregate demand, oil-specific (precautionary) demand
  - real activity: the index of Kilian (2009) as corrected in Kilian (2019) (FRED IGREA); price: refiner acquisition cost of imported crude, deflated by the US CPI
- Our sample starts in 1974 (first month of the price series), effective sample from January 1976, $T = 384$; extension to May 2026

Responses to oil supply and demand shocks



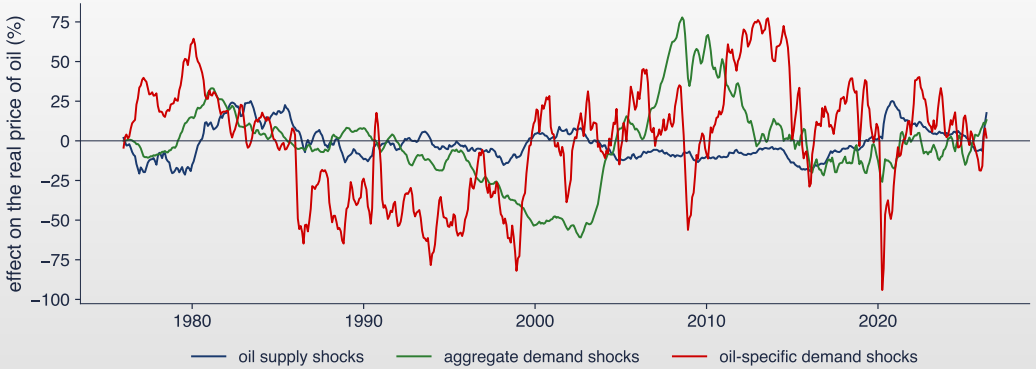
- One-standard-deviation shocks, production cumulated; 68% and 95% recursive-design wild-bootstrap bands Gonçalves and Kilian (2004); dashed: sample extended to 2026

ATS_ch3_oil_shocks

Interpreting the oil responses

- Supply shock: production falls by -1.5% on impact, the real price barely moves (-0.1% after 12 months, band $[-2.8, 2.2]$)
  - in the extended sample the effect is larger (2.0%): the post-2008 market reacts more to supply news
- Aggregate demand: a delayed, persistent rise of the price (4.8% after 12 months, band [1.5, 6.8])
- Oil-specific demand: an immediate jump (5.9%) that persists (7.1% after 12 months)
- At 60 months, supply shocks explain 1% of the price variance, aggregate demand 64%, oil-specific demand 35%: oil prices are mostly demand-driven

Historical decomposition of the real price of oil



⊡ Cumulative contribution of each structural shock, 1976–2026 (recursive VAR(24), extended sample); log points ×100

ATS_ch3_oil_shocks

Interpreting the historical decomposition

- 2003–mid-2008: aggregate demand adds 126 log points, supply -12: the boom was a global demand boom
- Mid-2008 to early 2009: oil-specific demand -82, aggregate demand -35; early 2020: oil-specific demand -103
- Mid-2021 to mid-2022: oil-specific demand 43, supply -2: the model reads the 2022 surge as precautionary demand
- Caveat: the zero restriction on the supply response is strong; Baumeister and Hamilton (2019) allow a small but non-zero elasticity and find a larger role for supply
  - a decomposition is only as good as the identifying assumption behind it

Blanchard and Quah: the long-run multiplier (1/2)

- For $y_t = (\Delta\text{output}_t, u_t)'$ the cumulative responses are
  - $\Theta(1) = \sum_h \Theta_h = (I - A_1 - \cdots - A_p)^{-1} B_0 = A(1)^{-1} B_0$
  - here u_t is the unemployment rate; $A(1) = I - A_1 - \cdots - A_p$: the VAR polynomial at 1; the first row of $\Theta(1)$ is the long-run effect on the *level* of output
- Restriction Blanchard and Quah (1989): demand shocks have no long-run effect on the level of output
  - $[\Theta(1)]_{12} = 0$
  - $[\cdot]_{12}$: the element in row 1 (output) and column 2 (the demand shock)

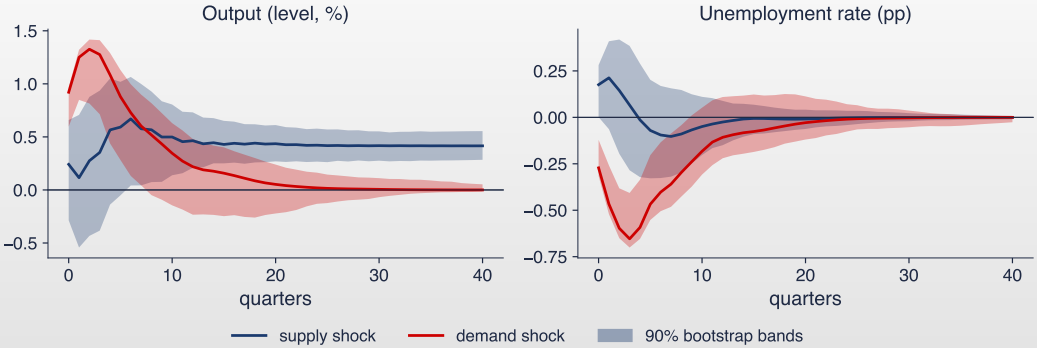
Blanchard and Quah: the long-run multiplier (2/2)

- Algorithm: $\Theta(1)\Theta(1)' = A(1)^{-1}\Sigma_u A(1)^{-1\prime}$ is known from the reduced form
  - take its lower-triangular Cholesky factor L (the zero in position $(1,2)$), then $B_0 = A(1)L$
  - the zero is placed on the long-run matrix, not on the impact matrix; derivation in the Appendix and in Seminar 3, A3
- Interpretation: "supply" = all shocks with permanent effects (technology, labour supply); Galí (1999) uses the same idea for technology shocks and hours

Case study: Blanchard and Quah (1989)

- The paper: US real GNP growth and the unemployment rate (men aged 20 and over), quarterly, VAR(8), 1950:2–1987:4 Blanchard and Quah (1989)
  - data treatment: separate means of output growth before and after 1973:4 (the productivity slowdown), a linear trend removed from unemployment
- Our replication: the same specification on today's vintage (FRED GNPC96, LNS14000025), $T = 151$; 90% residual-bootstrap bands
- Question: do demand shocks drive the business cycle, and how long do their effects last?

Supply and demand shocks identified by a long-run restriction



⊡ Output level (cumulated growth) and unemployment; one-standard-deviation shocks; 90% bootstrap bands

ATS_ch3_long_run

Interpreting the Blanchard–Quah responses

- Demand shock: output peaks at 1.33% after 2 quarters and returns to zero by construction; unemployment falls by -0.65 pp
- Supply shock: output rises permanently (0.42% after ten years, band [0.28, 0.55]), unemployment first *rises* (0.18 pp)
- Demand shocks explain 92% of the output-level variance at one year and 54% at ten years: in line with Blanchard and Quah (1989), demand dominates at business-cycle horizons
- Seminar 3, B5: without the break and the trend, the shares change dramatically

Reliability of long-run restrictions

- Faust and Leeper (1997): the long-run multiplier $A(1)^{-1}$ is estimated imprecisely from finite samples
  - a shock with a tiny but permanent effect is indistinguishable from a transitory one: inference can be arbitrarily distorted
  - the near-unit-root behaviour of unemployment makes the problem worse
- Results depend on the treatment of low frequencies: breaks in mean growth, trends, the lag length
  - Chapter 2 tests for such breaks; Chapter 4 treats cointegration, where long-run restrictions become restrictions on the VECM
- Use long-run restrictions with a sensitivity table, never alone

Recap: Long-run restrictions

- Restrict $\Theta(1) = A(1)^{-1}B_0$ instead of B_0: identification from permanence
- BQ: demand dominates business-cycle fluctuations in output, supply the long run
- Fragile: low-frequency treatment and the imprecise long-run multiplier drive the answer

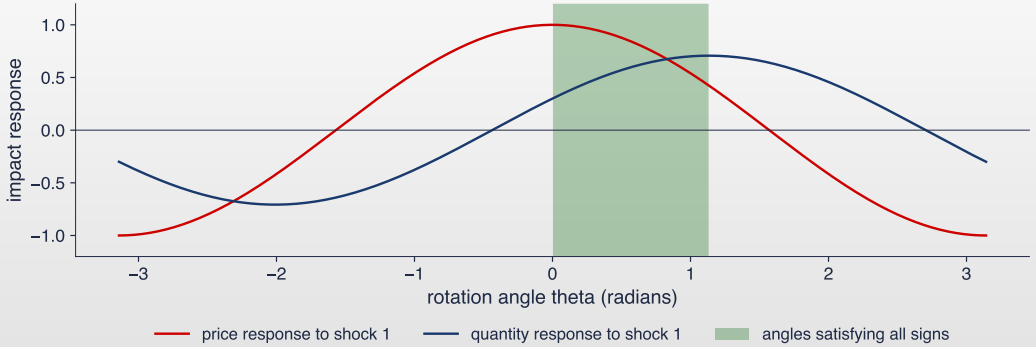
Identification by signs (1/2)

- Restrict only the **signs** of some responses for some horizons: a contractionary monetary shock raises the rate and lowers prices Uhlig (2005)
  - the response of interest (output) is left unrestricted: an "agnostic" procedure
- Algorithm Rubio-Ramírez, Waggoner and Zha (2010): draw a random rotation Q
  - draw X ($n \times n$) with i.i.d. $N(0,1)$ entries and compute its QR decomposition $X = QR$ (Q orthogonal, R upper triangular)
  - normalise the signs of $\mathrm{diag}(R)$ to be positive: then Q is uniform (Haar) on the set of orthogonal matrices

Identification by signs (2/2)

- Accept or reject the rotation
  - compute $\Theta_h = \Phi_h P Q$ and keep Q if all signs hold; repeat for draws of (A, Σ_u) from the posterior
  - zero restrictions together with signs need the importance-sampling algorithm of Arias, Rubio-Ramírez and Waggoner (2018)
- The result is a **set** of models consistent with the data and the restrictions, not a point

Sign restrictions in two dimensions



⊡ Price and quantity innovations with variances 1 and 0.5 and covariance 0.3; impact responses $P(\cos\theta, \sin\theta)'$ to shock 1 as the rotation angle varies; demand: $(+,+)$, supply: $(+,-)$

ATS_ch3_sign_restrictions

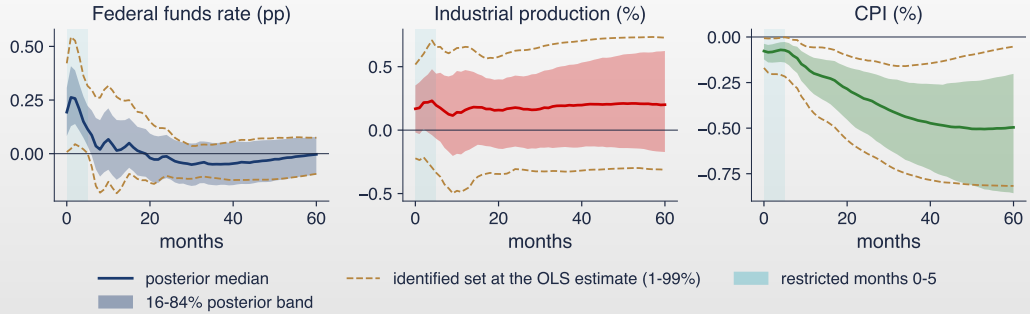
Interpreting the rotation set

- All signs hold for $\theta \in (0, 1.13)$, 18% of the circle: a whole interval of models is equally consistent with the data
- The impact effect of demand on quantity lies anywhere in $[0.30, 0.71]$: the **identified set**
- A uniform draw of θ puts a prior on this interval: the posterior inside the set is the prior, not information from data
  - this is the point of Baumeister and Hamilton (2015): Haar draws are informative about the responses

Case study: Uhlig (2005)

- The paper: six-variable monthly US VAR(12), no constant, 1965:1–2003:12; a contractionary shock raises the funds rate and lowers prices, commodity prices and nonborrowed reserves for months 0–5 Uhlig (2005)
  - output is not restricted: the question is its sign
- Our replication: the same restrictions and sample; IP and CPI replace the interpolated monthly GDP and deflator of the paper (data of Ramey (2016))
  - flat Normal–inverse-Wishart posterior, 1 000 draws, Haar impulse vectors; accepted: 985 draws; at the OLS estimate 4.1% of rotations satisfy the signs
- Question: does a contractionary monetary shock lower output when output is left free?

An agnostic monetary policy shock



One-standard-deviation shock; posterior median and 16–84% band; dashed: 1–99% range of the identified set at the OLS estimate; shaded: restricted months

ATS_ch3_sign_restrictions

Interpreting the agnostic responses

- Output after 12 months: median 0.16%, band $[-0.17, 0.47]$; 32% of draws negative: no evidence that output falls
- The identified set at the OLS estimate spans $[-0.47, 0.66]$: the data with the signs cannot sign output
- Prices fall persistently (-0.34% after 24 months) without a price puzzle, because the restriction rules it out
- Uhlig's conclusion, confirmed: neutrality of monetary policy shocks for output is not inconsistent with the data

Inference under set identification

- The posterior median is not a structural model: at each horizon it can come from a different rotation Fry and Pagan (2011)
  - report a single admissible model close to the median (median target) or, better, the whole set
- The Haar prior is informative about the responses Baumeister and Hamilton (2015); Giacomini and Kitagawa (2021) propose robust Bayes: report the range of posterior means over all priors on Q
  - the robust band is wider and honest: it separates what the data say from what the prior says
- Tighten the set with more information: more signs, zeros Arias, Rubio-Ramírez and Waggoner (2018), narrative events, elasticity bounds Baumeister and Hamilton (2019)

Narrative restrictions



Paul Volcker, Fed Chairman 1979–1987

Photo: Federal Reserve (2014); public domain; Wikimedia Commons

- Antolín-Díaz and Rubio-Ramírez (2018): restrict the structural shocks at specific dates, not the responses
  - October 1979 (the Volcker shift to reserve targeting) was a positive monetary policy shock
  - and it was the main contributor to the unexpected change of the funds rate that month
- A few credible events remove most of the rotations that sign restrictions alone leave
  - the likelihood must be reweighted by the probability of satisfying the narrative (importance sampling)
- With narrative restrictions output falls after a contractionary shock, unlike in the agnostic set

Identification through heteroskedasticity

- If the variances of the structural shocks change across regimes while B_0 stays fixed: $\Sigma_1 = B_0 B_0'$, $\Sigma_2 = B_0 \Lambda B_0'$, Λ diagonal Rigobon (2003)
  - Σ_1, Σ_2: the reduced-form covariance matrices of the two regimes; $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$: the relative variances of the shocks in regime 2
  - then $\Sigma_1^{-1}\Sigma_2 = B_0^{-1\prime}\Lambda B_0'$: the columns of B_0 follow from an eigendecomposition, unique if the λ_i are distinct
  - regimes: known dates (crises, policy announcements) or estimated (Markov switching, GARCH) Lanne and Lütkepohl (2008)
- Statistical identification gives shocks without names: labelling them still needs economics (signs)
  - the assumption that B_0 is the same in both regimes is strong and testable only partly
- Seminar 3, A8 recovers B_0 from two covariance matrices by hand; Chapter 7 develops regime switching

Recap: Sign restrictions and set identification

- Signs give a set of models; report the set, not only the posterior median
- Uhlig: with prices, rates and reserves restricted, output can go either way
- The Haar prior matters; robust Bayes and narrative restrictions sharpen honest conclusions
- Heteroskedasticity identifies statistically; economics still names the shocks

The proxy SVAR (1/2)

- An external variable z_t (the proxy), correlated with the shock of interest ε_{1t} and with no other shock
  - **relevance**: $\mathbb{E}z_t\varepsilon_{1t} = \alpha \neq 0$; **exogeneity**: $\mathbb{E}z_t\varepsilon_{jt} = 0$ for $j \neq 1$
- Then the covariance of the residuals with the proxy is proportional to the impact column b_1
  - $\mathbb{E}u_t z_t = B_0 \mathbb{E}\varepsilon_t z_t = \alpha b_1$ (derivation in the Appendix)
  - b_1: the first column of B_0, the impact effects of shock 1 on all n variables

The proxy SVAR (2/2)

- Normalise $b_{11} = 1$ (or a 25 bp rate increase) and estimate the relative impacts Stock and Watson (2012), Mertens and Ravn (2013)
  - $b_{i1}/b_{11} = \mathbb{E}(u_{it}z_t)/\mathbb{E}(u_{1t}z_t)$
  - b_{i1}: the impact on variable i; numerically a 2SLS regression of u_{it} on u_{1t} with instrument z_t
- Responses $\Theta_{h,\cdot 1} = \Phi_h b_1$: the first column of Θ_h
  - the proxy may be noisy, short or available only for a sub-sample
- Only one column of B_0 is identified, and no zero restriction is needed

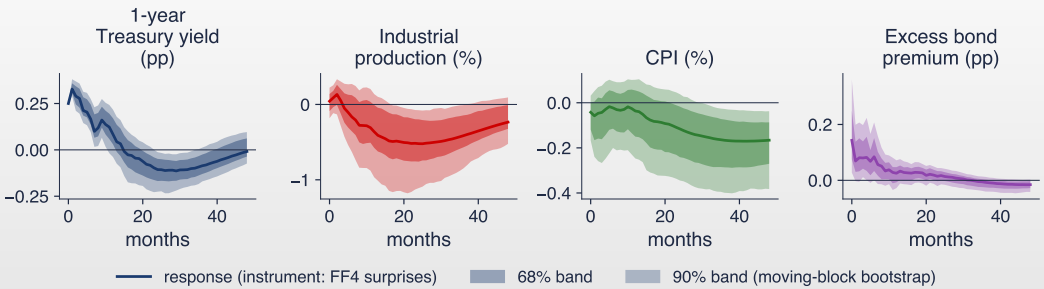
High-frequency identification of monetary policy

- Surprise = change of futures-implied rates in a narrow window (30 minutes) around an FOMC announcement Kuttner (2001), Gürkaynak, Sack and Swanson (2005)
  - within 30 minutes nothing else systematic happens: exogeneity by design
  - FF4: the three-month-ahead federal funds futures; it captures forward guidance as well as the current decision Gertler and Karadi (2015)
- Complications: the **information effect** Nakamura and Steinsson (2018): a tightening can signal good news about the economy
  - Jarociński and Karadi (2020) separate pure policy and information shocks with the co-movement of rates and stock prices (sign restrictions on surprises)
  - surprises are predictable from past macro news Bauer and Swanson (2023): orthogonalise them before use

Case study: Gertler and Karadi (2015)

- The paper: the 1-year Treasury yield as policy indicator (it captures forward guidance), log IP, log CPI and the excess bond premium of Gilchrist and Zakrajšek (2012); VAR(12), 1979:7–2012:6 Gertler and Karadi (2015)
  - instrument: FF4 surprises, monthly, 1991:1–2012:6; the residuals of the full sample are used for the proxy regression on the overlap
- Our replication: the same variables, sample and instrument (replication files of Ramey (2016)), $T = 384$, 258 months with the instrument
  - first stage: robust $F = 17.7$ (homoskedastic 21.9), $R^2 = 7.8\%$; 90% moving-block bootstrap bands Jentsch and Lunsford (2019)
- Question: does a policy surprise move credit spreads, and with what effect on activity?

A monetary policy shock identified with FF4 surprises



- Shock normalised to raise the 1-year yield by 25 bp on impact; 68% and 90% moving-block bootstrap bands

ATS_ch3_proxy_svar

Interpreting the proxy SVAR

- The excess bond premium jumps by 0.14 pp on impact (band [0.03, 0.35]): monetary policy works through credit costs, in line with Gertler and Karadi (2015)
- IP reaches -0.52% after 25 months (band $[-1.05, -0.12]$); CPI is -0.12% after two years (band $[-0.34, 0.04]$), with no price puzzle
- $F = 17.7$ passes the rule of thumb $F > 10$, but the price response is imprecise: weak-instrument-robust bands are wider Montiel Olea, Stock and Watson (2021)

Inference with an external instrument

- Weak instruments: when α is small, the ratio $\mathbb{E}(u_i z)/\mathbb{E}(u_1 z)$ is badly estimated and delta-method bands undercover
  - first-stage F: the F statistic of the regression of u_{1t} on z_t; it measures the strength of the instrument
- Remedies
  - Montiel Olea, Stock and Watson (2021): Anderson–Rubin-type confidence sets, valid for any strength; they can be unbounded
  - report the first-stage F with HAC variance; the rule $F > 10$ is a heuristic, not a test
- Bootstrap: the wild bootstrap of Mertens and Ravn (2013) is invalid for proxy SVARs; use the moving-block bootstrap of residuals and instrument jointly Jentsch and Lunsford (2019)
  - the block keeps the dependence between the squared shock and the instrument
- Seminar 3, A5 computes relative impacts and F by hand; C1 builds an Anderson–Rubin set for Romania

Recap: External instruments

- Relevance + exogeneity of a proxy identify one column of B_0, with no zeros
- High-frequency surprises are the best proxies we have; check the information effect and predictability
- GK replicated: credit spreads jump, output falls; $F = 17.7$
- Use weak-IV-robust sets and the block bootstrap

Delta method and its limits

- $\hat{\Theta}_h = g(\hat{\beta}, \hat{\sigma})$ is a smooth function of the VAR estimates: $\sqrt{T}(\hat{\Theta}_h - \Theta_h) \to N(0, G\Sigma G')$ Lütkepohl (2005)
  - $\hat{\beta}$: the VAR coefficients; $\hat{\sigma}$: the distinct elements of $\hat{\Sigma}_u$; Σ: their joint asymptotic covariance
  - $G = \partial g / \partial(\beta, \sigma)$: the Jacobian, in closed form; quick, but the bands are symmetric
- It fails where we need it most
  - near unit roots and at long horizons the distribution of $\hat{\Theta}_h$ is skewed and non-Normal
  - the covariance $G\Sigma G'$ can be singular (e.g. when a response is zero by construction)
  - OLS slope estimates are biased towards zero in small samples, so responses decay too fast
- Hence the bootstrap, and the bias correction

Bootstrap schemes for VARs

- **Residual (recursive-design)**: resample $\hat{u}_t$ i.i.d., rebuild y_t^* recursively from the estimated VAR, re-estimate, identify, repeat
  - valid with i.i.d. innovations; the percentile (or Hall) interval of the B responses gives the band
- **Wild**: $u_t^* = \eta_t \hat{u}_t$, $\eta_t = \pm 1$ with probability $1/2$: robust to conditional heteroskedasticity Gonçalves and Kilian (2004)
  - used for the oil VAR: commodity markets have volatility clustering
- **Moving block**: resample blocks of consecutive $(\hat{u}_t, z_t)$; needed whenever higher moments matter (proxy SVAR) Jentsch and Lunsford (2019)
  - block length of order $T^{1/4}$; Chapter 0 covered the choice

Kilian's bias-corrected bootstrap

- Kilian (1998): (1) bootstrap the bias $\hat{b} = \bar{A}^* - \hat{A}$; (2) correct $\tilde{A} = \hat{A} - \hat{b}$, shrinking the correction if $\tilde{A}$ is not stationary; (3) bootstrap again from $\tilde{A}$ and correct each replicate
  - $\hat{A}$: the OLS estimate of the VAR coefficients; $\bar{A}^*$: the mean of the bootstrap estimates; $\tilde{A}$: the bias-corrected estimate
  - the "bootstrap after bootstrap" recentres the distribution where OLS bias is large: persistent series, short samples
- Oil VAR: mean absolute bias 0.009 per coefficient; largest root 0.994 before, 0.9999 after the (shrunk) correction
  - 12-month response of the price to oil-specific demand: 7.11% (OLS), 7.34% (corrected), band [3.4, 9.1]; to aggregate demand: 4.83% and 4.95%
- The correction makes responses more persistent; when the VAR in levels has a root above one, the rule of Kilian leaves it uncorrected (Romanian VAR, Section 9)

Pointwise and joint bands

- A pointwise 90% band covers Θ_h at each h separately; the whole path lies inside it with much lower probability
  - questions about the *shape* ("output falls for two years") need simultaneous bands Montiel Olea and Plagborg-Møller (2019)
  - sup-t band: scale the pointwise standard errors by the $1-\alpha$ quantile of $\max_h |t_h|$ from the bootstrap
  - t_h: the bootstrap response at horizon h minus the estimate, over its standard error; $\max_h |t_h|$: the largest deviation along the whole path
- Bayesian bands (posterior quantiles) answer a different question: probability statements given the prior
  - Chapter 5 builds them with Minnesota and Normal–inverse-Wishart priors; with set identification they mix prior and data (previous section)
- Always state: pointwise or joint, frequentist or Bayesian, the level (68%, 90%, 95%)

Recap: Inference for impulse responses

- Delta-method bands are symmetric and fail near unit roots
- Residual, wild and block bootstraps match i.i.d., heteroskedastic and proxy settings
- Kilian's correction removes small-sample bias; it is not applied to explosive estimates
- Shape questions need joint bands; Bayesian bands come in Chapter 5

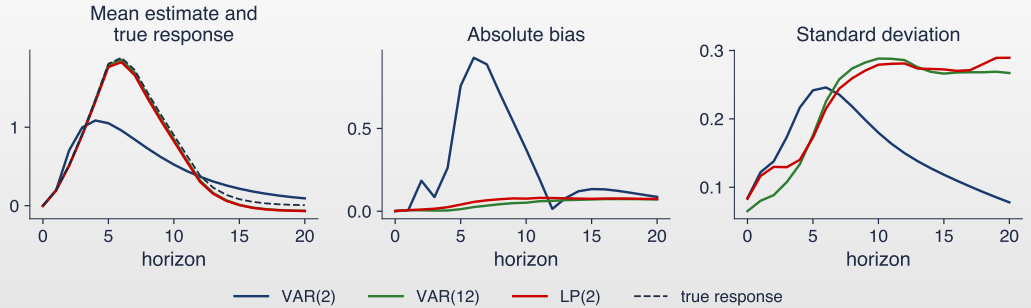
Local projections

- Jordà (2005): one regression per horizon, $y_{i,t+h} = \mu_h + \beta_h x_t + \gamma_h' w_t + \xi_{t+h}$, $h = 0, 1, \ldots, H$
  - $y_{i,t+h}$: variable i, h periods after the shock; μ_h: an intercept; ξ_{t+h}: the projection error; H: the longest horizon
  - x_t: the shock (or the policy variable with controls w_t that make it as good as random); $\hat{\beta}_h$ is the impulse response
  - no iteration of a model: a misspecified dynamic does not compound across horizons
- The error ξ_{t+h} is serially correlated (an MA($h-1$) at least): HAC standard errors with $h+1$ lags, or
  - **lag augmentation** Montiel Olea and Plagborg-Møller (2021): add one more lag of the controls and use Eicker–Huber–White errors; valid uniformly over persistence, even near a unit root
- Flexible: nonlinear terms, state dependence, panel data, cumulative outcomes all enter as regressors

LP and VAR estimate the same responses

- Plagborg-Møller and Wolf (2021): in population, an LP with p lags of all variables as controls and a VAR(p) with the same ordering give *identical* responses up to horizon p (Appendix)
  - with unrestricted lags ($p \to \infty$) they coincide at all horizons: the choice is about estimation, not identification
  - any VAR identification scheme (recursive, external instrument) has an LP counterpart, and vice versa
- In finite samples they differ: the VAR extrapolates the first p autocovariances, the LP uses the raw h-step covariance
  - the next slide quantifies this

Bias and variance: LP against VAR (simulation)



- DGP: $x_t = 0.5x_{t-1} + e_t$, $y_t = 0.6y_{t-1} + \sum_{j=0}^{11} \psi_j e_{t-j} + u_t$ (hump-shaped ψ_j), $T = 240$, 500 replications; e_t, u_t: independent white noises; a VAR(2) is misspecified

ATS_ch3_lp_vs_var

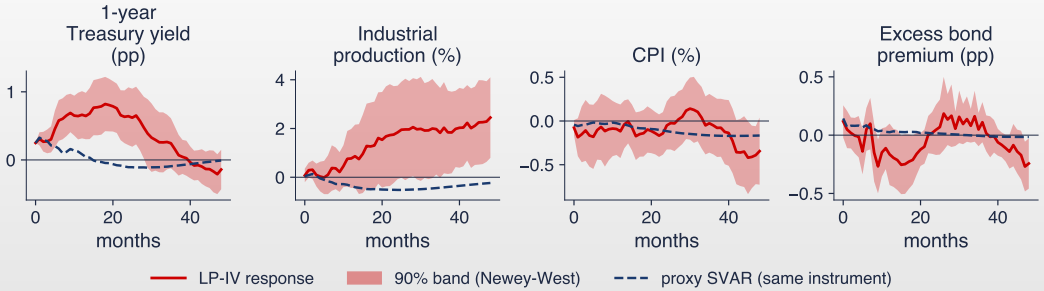
Interpreting the bias–variance trade-off

- At $h = 8$ (true response 1.44): VAR(2) bias 0.71, LP(2) bias 0.07; RMSE 0.74 and 0.27
  - where the short VAR misses the hump, its bias dominates the error
- At $h = 16$: standard deviation 0.11 (VAR(2)) against 0.27 (LP(2)); RMSE 0.17 against 0.28: the biased VAR wins
- The general lesson of Li, Plagborg-Møller and Wolf (2024) (thousands of DGPs): LP has low bias and high variance; VAR (and shrinkage VARs) have lower MSE unless the analyst cares almost only about bias
- A long VAR(12) behaves like LP: few lags = more bias, less variance

LP-IV

- Stock and Watson (2018): $y_{i,t+h} = \theta_{h,i}\, y_{1t} + \gamma' w_t + \xi_{t+h}$, with y_{1t} (the policy rate) instrumented by z_t
  - $\theta_{h,i}$ = response of y_i after h periods to a shock that raises y_1 by one unit on impact
  - conditions: relevance, contemporaneous exogeneity and **lead-lag exogeneity** $\mathbb{E} z_t \varepsilon_{t+j} = 0$ for $j \neq 0$, given w_t
  - lead-lag exogeneity: the instrument is unrelated to past and future shocks, not only to current ones
- Same identifying information as the proxy SVAR; different estimator
  - the proxy SVAR needs invertibility of the shock; LP-IV does not, but pays in variance
  - cumulative LP-IV gives multipliers: $\sum_{j \leq h} y_{t+j}$ on $\sum_{j \leq h} g_{t+j}$ instrumented by z_t Ramey and Zubairy (2018); y: output, g: government spending

LP-IV and the proxy SVAR with the FF4 instrument



- LP-IV with FF4 as instrument for the 1-year yield, 2 lags of all variables and of the instrument as controls, origins 1990:1–2012:6 ($T = 270$) Ramey (2016); dashed: the proxy SVAR

ATS_ch3_lp_vs_var

Interpreting LP-IV against the proxy SVAR

- Impact first stage $F = 21.5$; after 24 months IP: LP-IV 1.84% (SE 1.17), proxy SVAR -0.52%
- CPI after 24 months: LP-IV -0.05% (SE 0.17), proxy SVAR -0.12%: the LP bands contain the VAR path almost everywhere
- LP-IV implies a much more persistent rise of the yield: the two estimators answer slightly different questions when the policy path differs
  - the positive LP output response, noted by Ramey (2016), is imprecise and sensitive to the sample: do not read it as expansionary tightening

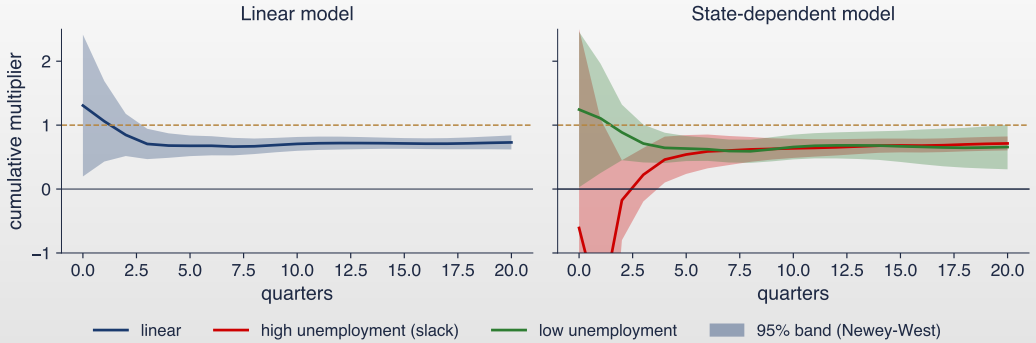
State-dependent local projections (1/2)

- Interact everything with the state indicator of the previous period, I_{t-1}:
  - $y_{t+h} = I_{t-1}[\alpha_{A,h} + \beta_{A,h} x_t + \ldots] + (1 - I_{t-1})[\alpha_{B,h} + \beta_{B,h} x_t + \ldots] + \xi_{t+h}$
  - $I_{t-1} = 1$ in state A (e.g. slack), 0 in state B; $\beta_{A,h}$, $\beta_{B,h}$: the responses at horizon h in each state; α: state-specific intercepts
- What LP measures here
  - the state may change after the shock: LP measures the average response given the initial state, including transitions
  - no need to model the state dynamics, unlike a threshold or a smooth-transition VAR (Chapter 2)

State-dependent local projections (2/2)

- Case study Ramey and Zubairy (2018): are government spending multipliers larger in slack times?
  - US quarterly data 1889–2015; military news shock (Ramey) scaled by potential GDP
  - slack = unemployment $\geq 6.5\%$ in $t-1$ (36% of quarters)
  - cumulative LP-IV of output on government spending, 4 lags of news, output and spending, Newey–West with $h+1$ lags

Government spending multipliers by state of the economy



- Cumulative multiplier $\sum_{j \le h} \Delta y_{t+j} / \sum_{j \le h} \Delta g_{t+j}$ by horizon; 95% Newey–West bands; dashed: multiplier of one

ATS_ch3_state_dependent_lp

Interpreting the state-dependent multipliers

- Linear: 0.67 at two years (SE 0.06) and 0.71 at four years (SE 0.04)
- High unemployment: 0.62 and 0.68; low unemployment: 0.59 and 0.66
- In line with Ramey and Zubairy (2018): multipliers below one in both states and no significant difference between them
- The case for LP here: a VAR would impose the same dynamics in both states unless the state transitions are modelled

Recap: Local projections

- LP = one regression per horizon; HAC or lag-augmented inference
- LP and VAR target the same responses; finite samples trade bias for variance
- LP-IV uses external instruments without invertibility, at a cost in precision
- State dependence is natural in LP: RZ find multipliers below one in slack and in normal times

Monetary policy in a small open economy

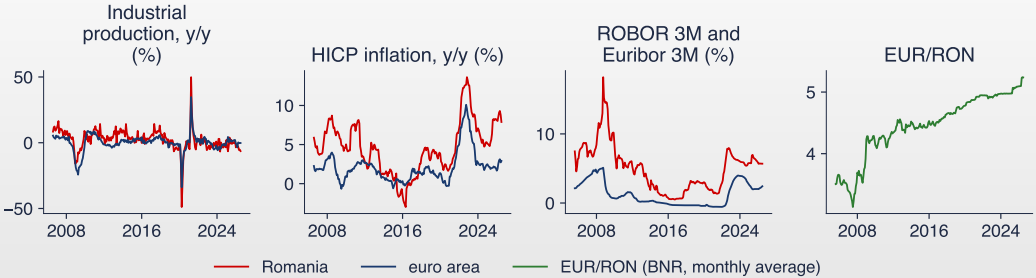


The National Bank of Romania, Bucharest

Photo: Ștefan Jurcă (2015); CC BY 2.0; Wikimedia Commons

- The BNR targets inflation since August 2005 (target 2.5% $\pm$ 1 pp since 2013) under a managed float of the leu
  - the interbank rate ROBOR 3M transmits the policy rate and liquidity conditions; we use it as the policy indicator
- Identification problems specific to Romania
  - short sample (20 years), structural change (2008–2009, 2020, 2022), administered prices and excise changes
  - external shocks dominate: the euro area takes most Romanian exports; ECB policy moves Romanian rates and the leu
- Our approach: a transparent recursive VAR with the euro area first, then an external ECB shock with LP

Romanian and euro-area data



- Monthly, August 2005–July 2026: IP (SCA), HICP, 3-month money-market rates (Eurostat), EUR/RON (BNR reference rate, monthly average)

ATS_ch3_romania

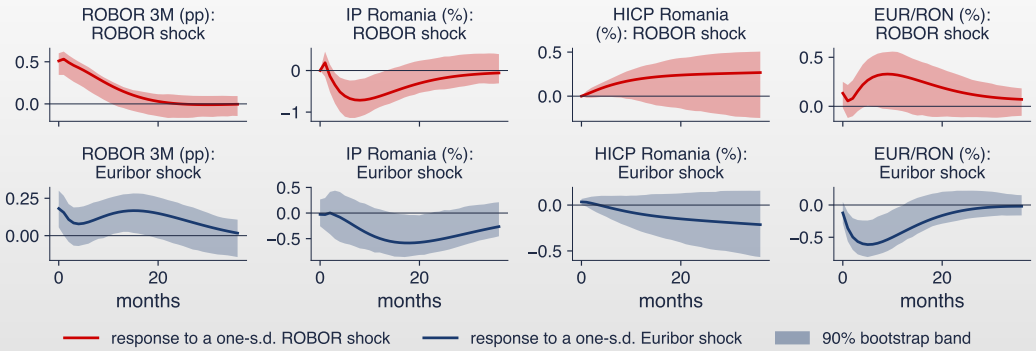
Interpreting the Romanian data

- HICP inflation peaked at 13.6% in November 2022 and is 7.9% in the last month; ROBOR 3M reached 18.2% in the 2008 liquidity squeeze
- EUR/RON rose from 3.51 to 5.24: a trend depreciation with long managed-float plateaus
- Romanian and euro-area IP co-move closely, including the 2020 collapse: a strong case for an external block
- Sharp outliers (2008, 2020, 2022) will dominate the estimates of a small VAR: check robustness without them

A VAR for Romania with a euro-area block

- y_t = (EA IP, EA HICP, Euribor 3M, RO IP, RO HICP, ROBOR 3M, EUR/RON), logs $\times 100$ except rates; constant and monthly dummies (HICP is not seasonally adjusted)
  - lag length: BIC and HQ choose 2, AIC 6; we use 2 ($T = 250$); largest root 1.006: levels, near-unit-root, no bias correction
- Identification (Cholesky, in this order): the euro area does not react to Romania within the month; ROBOR reacts to Romanian activity and prices within the month; the exchange rate reacts to everything
  - the zero restriction on the euro area is credible (Romania is small); the position of ROBOR before EUR/RON is debatable
  - a block-exogenous VAR would also set the lagged Romanian effects on the euro area to zero (Seminar 3, B2 discusses orderings)

Responses to a domestic and a euro-area rate shock



- VAR(2), Cholesky with the euro-area block first; one-standard-deviation shocks (ROBOR 0.51 pp, Euribor 0.09 pp); 90% residual-bootstrap bands

ATS_ch3_romania

Interpreting the Romanian responses

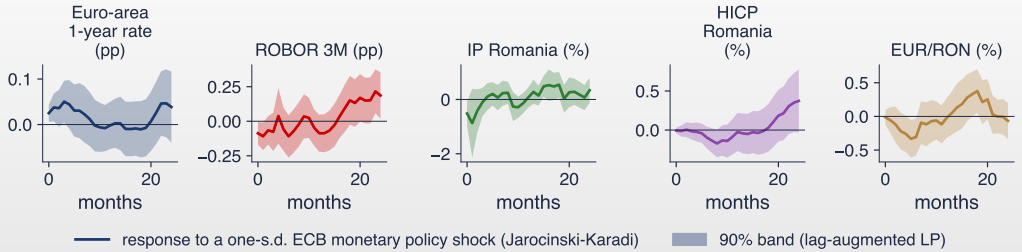
- ROBOR shock: IP -0.62% after 12 months (band $[-0.88, -0.06]$); HICP 0.20% (band $[-0.06, 0.34]$); EUR/RON 0.31%
  - a price puzzle and an exchange-rate puzzle (depreciation after a tightening): the ROBOR innovation still contains the BNR's reaction to inflation and to pressure on the leu
- Euribor shock: Romanian IP -0.50% after 12 months (band $[-0.77, 0.07]$); HICP -0.10%; ROBOR 0.16 pp
- Honest summary: the recursive VAR identifies the euro-area block credibly and the domestic policy shock poorly

The euro-area share of Romanian fluctuations

Variable	EA shocks, 12 m	EA shocks, 36 m	ROBOR shock, 12 m	own shock, 12 m
IP Romania	46%	37%	8%	41%
HICP Romania	44%	55%	3%	42%
ROBOR 3M	30%	39%	59%	–
EUR/RON	25%	23%	6%	58%

- Forecast error variance decomposition of the recursive VAR(2); EA shocks = the three euro-area shocks together
- Euro-area shocks explain 46% of Romanian IP at one year and 55% of HICP at three years; the domestic rate shock explains less than a tenth of either
- The shares are as credible as the ordering; with a block-exogenous VAR they are a lower bound for external influence

ECB policy shocks and Romania: local projections



- Shock: the pure monetary policy shock of Jarociński and Karadi (2020) (authors' update, one s.d. = 3.2 bp); 3 lags of all variables and of the shock (lag augmentation), monthly dummies; origins 2005:8–2025:10

ATS_ch3_romania

Interpreting the spillover projections

- Relevance first: the euro-area 1-year rate rises by 0.03 pp on impact (SE 0.01); first-stage $F = 4.8$ (impact) and 4.0 (one month)
  - monthly averages dilute an announcement-day surprise: the shock is a weak instrument for monthly euro-area rates
- Romanian HICP: -0.02% after 12 months (SE 0.15), 0.37% after 24 (SE 0.25); EUR/RON -0.31% after 6 months (SE 0.17)
- Honest conclusion: with a weak external shock and 20 years of data, the spillovers to Romanian prices are not determined; a weak-IV-robust set is wide (Seminar 3, C1)

Conclusions for Romania

- Robust findings
  - euro-area shocks account for a large share of Romanian output and price fluctuations
  - a euro-area tightening lowers Romanian IP within a year in the recursive VAR
- Fragile findings
  - the effect of a domestic ROBOR shock on prices and the leu: puzzles signal an identification failure, not a fact about Romania
  - the size of ECB spillovers to Romanian inflation: the instrument is weak at a monthly frequency
- Next steps (projects)
  - BNR policy-meeting surprises from ROBOR or FRA quotes around announcements, used as a proxy
  - sign restrictions (a tightening appreciates the leu and lowers prices) and a Bayesian VAR with shrinkage (Chapter 5)

Recap: Romania and euro-area spillovers

- Put the large neighbour first: the euro-area block is credibly exogenous within the month
- A recursive ROBOR shock produces price and exchange-rate puzzles
- External ECB shocks are weak instruments at a monthly frequency: report the uncertainty, do not hide it

An open question

- How strongly does ECB monetary policy pass through to Romanian inflation, and has the pass-through changed since 2022?
  - formal: H_0: $\theta_{12} = 0$, the 12-month response of Romanian HICP to a 25 bp euro-area rate shock, tested by an Anderson–Rubin set that is valid with weak instruments
  - falsified by a bounded 90% set that excludes zero for a pre-registered shock, horizon and sample
- Why it matters: Romania plans to join the euro area; the BNR needs to know how much of its inflation is imported
  - literature to start from: Jarociński and Karadi (2020), Stock and Watson (2018), Montiel Olea, Stock and Watson (2021), Plagborg-Møller and Wolf (2021), Li, Plagborg-Møller and Wolf (2024)

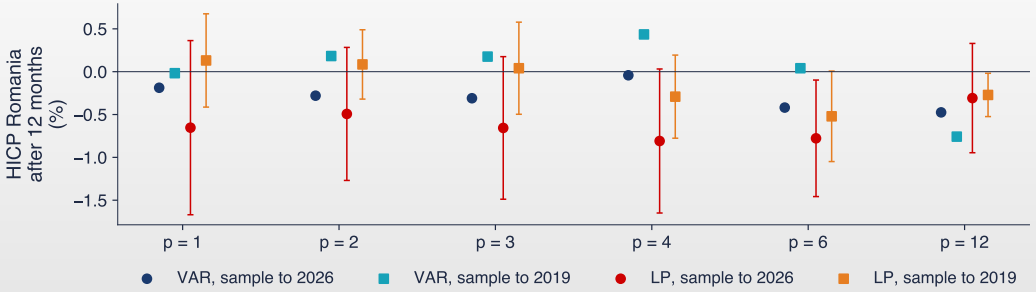
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed papers that estimate euro-area monetary spillovers to Romania or Central and Eastern Europe with high-frequency shocks; give DOIs.` Then check every DOI on Crossref
  - **hypothesis**: `Propose three channels (trade, exchange rate, bank funding) and one observable implication of each.`
  - **code and replication**: ask for an LP-IV function, then reproduce a known number first (the first-stage F of this lecture)
  - **robustness and critique**: `Act as a hostile referee: list every way this identification of ECB spillovers could fail.`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

Required checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The shock is dated correctly: announcement-day surprises are summed within the month, never shifted forward
- Relevance is reported (first-stage F with HAC variance) before any response is interpreted
- Lags, horizons, sample and the treatment of 2020 are fixed before the results; all variants are reported
- An AI summary of a "significant spillover" is checked against the confidence set, not against the point estimate

Mini-case: sensitivity of the HICP response to the specification



- Response of Romanian HICP after 12 months to a 25 bp Euribor shock: recursive VAR and recursive LP (90% Newey–West bars), lag length p, samples ending in 2019 and in 2026
- 24 estimates range from -0.81% to 0.44%, 7 of them positive: the sign depends on the lag length and on the post-2020 data; an AI assistant that reports one of them "with confidence" is wrong

ATS_ch3_romania

Project idea

- **Euro-area monetary spillovers to Romania**: replicate first, then extend
  - replicate: the proxy SVAR of Gertler and Karadi (2015) (this lecture: $F = 17.7$, the bond-premium jump) and the Romanian LP of this lecture
  - extend: daily ECB surprises aggregated to monthly with the Gertler–Karadi averaging; a quarterly GDP version; Anderson–Rubin sets; sign restrictions on the surprises (Jarociński and Karadi (2020))
  - pre-register: shock, outcomes, horizons 0–24, lag rule, the 2020 treatment and the Holm correction across outcomes
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- A structural VAR is a reduced form plus an identifying assumption; the assumption is the economics
- Zeros (short or long run), signs, narratives, heteroskedasticity and instruments each buy identification with a different assumption
- Set identification is honest: report the set and the role of the prior
- Inference: bootstrap matched to the setting, bias correction, joint bands, weak-IV-robust sets
- LP and VAR estimate the same responses; choose by bias and variance, and use LP for state dependence

Self-assessment

Questions

- How many restrictions identify a five-variable SVAR exactly?
- Why does $\mathbb{E} u_t z_t$ identify the impact column of a proxy SVAR?
- What does the posterior median of a sign-restricted response fail to represent?
- When does a VAR beat LP in mean squared error?
- Why is the wild bootstrap invalid for a proxy SVAR?

Next:

- Cointegration revisited: VECM, ARDL and panel data
- Long-run restrictions become restrictions on cointegrating vectors and on the common trends

Appendix: the long-run restriction in closed form

- Stable VAR: $\sum_{h\geq 0}\Phi_h = A(1)^{-1}$, so the long-run response matrix is $\Theta(1) = A(1)^{-1}B_0$
- $\Theta(1)\Theta(1)' = A(1)^{-1}B_0B_0'A(1)^{-1\prime} = A(1)^{-1}\Sigma_u A(1)^{-1\prime}$, known from the reduced form
- $[\Theta(1)]_{12} = 0$ makes $\Theta(1)$ lower triangular: $\Theta(1) = \mathrm{chol}(A(1)^{-1}\Sigma_u A(1)^{-1\prime})$, unique with a positive diagonal
- Hence $B_0 = A(1)\Theta(1)$; check $B_0B_0' = \Sigma_u$; signs are then normalised (supply raises output in the long run)

Back

Appendix: identifying the column b_1 with a proxy

- $u_t = B_0 \varepsilon_t = b_1 \varepsilon_{1t} + \sum_{j \geq 2} b_j \varepsilon_{jt}$
- $\mathbb{E} u_t z_t = b_1 \mathbb{E} \varepsilon_{1t} z_t + \sum_{j \geq 2} b_j \mathbb{E} \varepsilon_{jt} z_t = \alpha b_1$ by exogeneity
- Relative impacts $b_{i1}/b_{11} = \mathbb{E} u_{it} z_t / \mathbb{E} u_{1t} z_t$ need $\alpha \neq 0$ (relevance); the scale of b_1 follows from $b_1' \Sigma_u^{-1} b_1 = 1$

Back

Appendix: LP and VAR agree up to horizon p

- Let $w_t = (y_{t-1}, \ldots, y_{t-p})$ and the shock be the innovation $\tilde{x}_t = x_t - \mathrm{proj}(x_t \mid w_t)$
- LP coefficient: $\beta_h = \mathrm{Cov}(y_{t+h}, \tilde{x}_t)/\mathrm{Var}(\tilde{x}_t)$ (Frisch–Waugh)
- The VAR(p) reproduces the autocovariances $\Gamma_0, \ldots, \Gamma_p$ exactly; for $h \leq p$, $\mathrm{Cov}(y_{t+h}, \tilde{x}_t)$ depends only on them, so both give the same β_h Plagborg-Møller and Wolf (2021)
- For $h > p$ the VAR extrapolates its own autocovariances: the source of the bias of a short VAR, and of the lower variance

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